Assignment 4 — Write-up

DSC 190/291 — Learning Theory Student: Zeyu Bian

Part A — Sparse Linear Predictors and Model Selection

Throughout, $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{-1, +1\}$, $\mathcal{H}_k = \{x \mapsto \mathrm{sign}(\langle w, x\rangle) : \|w\|_0 \le k\}$, $\mathcal{H} = \bigcup_{k=1}^d \mathcal{H}_k$, and $S \sim \mathcal{D}^n$.

A.1 VC dimension through supports

For a fixed support $I \subseteq [d]$ with $|I| = k$, write

$$\mathcal{H}_I \;=\; \{x \mapsto \mathrm{sign}(\langle w, x\rangle) : \mathrm{supp}(w) \subseteq I\}.$$

The map $w \mapsto w|_I$ identifies $\mathcal{H}_I$ with the class of homogeneous halfspaces in $\mathbb{R}^k$, applied to the coordinates of x indexed by I. Homogeneous halfspaces in $\mathbb{R}^k$ have VC dimension k, so by Sauer–Shelah

$$\Gamma_{\mathcal{H}_I}(n) \;\le\; \left(\tfrac{en}{k}\right)^k \qquad (n \ge k).$$

Since $\mathcal{H}_k = \bigcup_{|I|=k} \mathcal{H}_I$, a union bound on dichotomies gives

$$\Gamma_{\mathcal{H}_k}(n) \;\le\; \binom{d}{k} \left(\tfrac{en}{k}\right)^k \;\le\; \left(\tfrac{ed}{k}\right)^k \left(\tfrac{en}{k}\right)^k \;=\; \left(\tfrac{e^2 dn}{k^2}\right)^k.$$

If $\mathrm{VCdim}(\mathcal{H}_k) = m$, then $2^m \le \Gamma_{\mathcal{H}_k}(m) \le (e^2 dm/k^2)^k$, so $m \le k \log_2(e^2 dm/k^2)$. A standard lemma (e.g. Shalev-Shwartz–Ben-David, Lemma A.2) converts $m \le k \log_2(am/k)$ into $m \le 2k \log_2(2ak/k) = O(k \log(ed/k))$. Hence

$$\boxed{\mathrm{VCdim}(\mathcal{H}_k) \;=\; O\big(k \log(ed/k)\big).}$$

Lower-bound sanity check. $\mathcal{H}_k$ contains $\mathcal{H}_I$ for any single I, and $\mathrm{VCdim}(\mathcal{H}_I) = k$; so $\mathrm{VCdim}(\mathcal{H}_k) \ge k$. The upper bound is tight up to the $\log(ed/k)$ factor, which is the price of choosing the support.

A.2 Penalty-based SRM

Class-level SRM theorem (from lecture). Let $\mathcal{H} = \bigcup_{k=1}^d \mathcal{H}_k$ with $p_k > 0$, $\sum_k p_k \le 1$. Suppose for each k there is a uniform-convergence rate $\varepsilon_k(n, \delta)$ such that, with probability $\ge 1 - \delta$ over S, $\sup_{h \in \mathcal{H}_k} |L_S(h) - L_{\mathcal{D}}(h)| \le \varepsilon_k(n, \delta)$. Define $\mathrm{pen}(k, n, \delta) = \varepsilon_k(n, p_k \delta)$ and $\hat{h} \in \arg\min_{k, h \in \mathcal{H}_k} \{L_S(h) + \mathrm{pen}(k, n, \delta)\}$. Then with probability $\ge 1 - \delta$,

$$L_{\mathcal{D}}(\hat{h}) \;\le\; \inf_{k,\; h \in \mathcal{H}_k} \{L_{\mathcal{D}}(h) + 2\,\mathrm{pen}(k, n, \delta)\}.$$

Instantiation for $\mathcal{H}_k$. By A.1, $\mathrm{VCdim}(\mathcal{H}_k) = d_k = O(k \log(ed/k))$. The VC uniform-convergence theorem (e.g. Vapnik–Chervonenkis / Massart) gives

$$\varepsilon_k(n, \delta) \;\le\; C_0 \sqrt{\frac{d_k + \log(1/\delta)}{n}} \;\le\; C_1 \sqrt{\frac{k \log(ed/k) + \log(1/\delta)}{n}}.$$

Apply with confidence parameter $p_k\delta$:

$$\mathrm{pen}(k,n,\delta) \;\le\; C\sqrt{\frac{k\log(ed/k)+\log(1/p_k)+\log(1/\delta)}{n}}.$$

By the SRM theorem, with probability $\ge 1-\delta$,

$$\boxed{L_{\mathcal{D}}(\hat{h}) \;\le\; \inf_{1\le k\le d,\ h\in\mathcal{H}_k}\left\{L_{\mathcal{D}}(h)+C\sqrt{\frac{k\log(ed/k)+\log(1/p_k)+\log(1/\delta)}{n}}\right\}.}$$

Interpretation of each term.

- $k\log(ed/k)$ is the **support cost**: it is the VC dimension of $\mathcal{H}_k$, and it pays for uniform convergence *within* the level k — i.e. for the freedom to choose which k coordinates carry weight and which halfspace to put on them.
- $\log(1/p_k)$ is the **sparsity-level cost**: it is the price of the union bound across the d sparsity levels. With $p_k \propto 1/k^2$ we have $\log(1/p_k) = 2\log k + O(1)$, much smaller than $\log(d)$ for the levels we actually care about.
- $\log(1/\delta)$ is the usual confidence cost.

A.3 Comparison with dense halfspaces

Suppose w^* with $\|w^*\|_0 \le s$ and $L_{\mathcal{D}}(w^*) \le \eta$. Plugging $h =$ this w^* and $k = s$ into the SRM bound and forcing the penalty $\le \varepsilon$:

$$n \;\ge\; C\,\frac{s\log(ed/s)+\log(1/p_s)+\log(1/\delta)}{\varepsilon^2} \;=\; O\!\left(\frac{s\log(ed/s)+\log(1/\delta)}{\varepsilon^2}\right)$$

(using $p_s \propto 1/s^2$, so $\log(1/p_s) = O(\log s)$, absorbed into $\log(ed/s)$).

Dense halfspaces in $\mathbb{R}^d$ have VC dimension d, so the corresponding sample complexity is

$$n_{\text{dense}} \;=\; O\!\left(\frac{d+\log(1/\delta)}{\varepsilon^2}\right).$$

When sparse is smaller. Sparse beats dense when $s\log(ed/s) \ll d$, i.e. roughly when $s = o(d/\log d)$. For $s = d$ the bounds are identical up to a log factor; for $s = \Theta(\log d)$ the sparse bound is exponentially better in d.

A.4 Validation as model selection

Split S evenly into S_1 (training, size $m = n/2$) and S_2 (validation, size m). Define

$$w_k \in \arg\min_{\|w\|_0\le k} L_{S_1}(w), \qquad \hat{k} \in \arg\min_{1\le k\le d} L_{S_2}(w_k).$$

Step 1 (validation step). Condition on S_1. The list $w_1,\ldots,w_d$ is then a *fixed* (i.e. S_2-independent) set of d classifiers. Apply Hoeffding plus a union bound over these d candidates on the independent sample S_2 of size $m = n/2$: with probability $\ge 1-\delta/2$, for all $k \in \{1,\ldots,d\}$,

$$\left|L_{S_2}(w_k) - L_{\mathcal{D}}(w_k)\right| \;\le\; \sqrt{\frac{\log(4d/\delta)}{2m}} \;\le\; C_2\sqrt{\frac{\log(d/\delta)}{n}}.$$

Therefore, on this event,

$$L_{\mathcal{D}}(w_{\hat{k}}) \;\le\; L_{S_2}(w_{\hat{k}}) + C_2\sqrt{\tfrac{\log(d/\delta)}{n}} \;\le\; L_{S_2}(w_k) + C_2\sqrt{\tfrac{\log(d/\delta)}{n}} \quad \forall k,$$

and combining the two Hoeffding inequalities,

$$L_{\mathcal{D}}(w_{\hat{k}}) \;\le\; \min_{1\le k\le d} L_{\mathcal{D}}(w_k) + 2C_2\sqrt{\tfrac{\log(d/\delta)}{n}}. \quad (\star)$$

Step 2 (training step). For each fixed k, w_k minimizes empirical risk over $\mathcal{H}_k$ on S_1. By the VC uniform-convergence bound applied to $\mathcal{H}_k$ with confidence $\delta/(2d)$ and sample size $m = n/2$, plus a union bound over k, with probability $\ge 1-\delta/2$, for every k and every w with $\|w\|_0 \le k$,

$$L_{\mathcal{D}}(w_k) \;\le\; L_{\mathcal{D}}(w) + C_3\sqrt{\frac{k\log(ed/k) + \log(d/\delta)}{n}}. \quad (\star\star)$$

Combine. Union bound the two events. With probability $\ge 1-\delta$, taking the infimum over k and w in $(\star)$ via $(\star\star)$,

$$\boxed{L_{\mathcal{D}}(w_{\hat{k}}) \;\le\; \inf_{1\le k\le d,\ \|w\|_0\le k} \left\{ L_{\mathcal{D}}(w) + C\sqrt{\frac{k\log(ed/k) + \log(d/\delta)}{n}} \right\}.}$$

Comparison with penalty SRM.

	Penalty-SRM	Validation
Sample used to *fit* candidates	all n	$n/2$
Sample used to *choose* k	the same n	a separate, independent $n/2$
Complexity term for support	$k\log(ed/k)$	$k\log(ed/k)$
Complexity term for choosing k	$\log(1/p_k) = O(\log k)$	$\log d$

Validation pays a constant factor of ~ 2 in n (the split) and replaces the prior-dependent $\log(1/p_k)$ by $\log d$. In exchange, the validation step is purely finite-class and never invokes the (in general computationally hard) penalty.

A.5 Computation

Brute-force algorithm (realizable case). Enumerate every support $I \subseteq [d]$ with $|I| = k$. For each I, solve the homogeneous-halfspace feasibility problem

$$\text{find } w \in \mathbb{R}^I : \quad y_i\langle w, x_i|_I\rangle \ge 0 \quad (i = 1, \ldots, n)$$

(strict-inequality / margin form, e.g. by LP or Perceptron). This is solvable in $\mathrm{poly}(k, n)$ time. If any support yields a feasible w, output it; the realizable assumption guarantees at least one such I exists.

Runtime. $\binom{d}{k} \cdot \mathrm{poly}(k, n) \le (ed/k)^k \cdot \mathrm{poly}(k, n)$. This is polynomial in d iff $k = O(1)$ (a constant). For $k = \Theta(\log d)$ already, it is $d^{\Theta(\log d)}$, superpolynomial.

Statistical vs computational complexity. Part A.3 shows that $n = \widetilde{O}(s/\varepsilon^2)$ samples suffice to learn an s-sparse halfspace, with *no* explicit dependence on d beyond a log. But the brute-force algorithm needs d^s time. So small sample complexity does not buy small runtime: choosing the support is a combinatorial problem whose decision version (PAC-learning sparse halfspaces under adversarial noise) is NP-hard in general. This is exactly the Week 4 picture — VC theory says "the information is in the sample"; computation says "extracting it can still be hard."

Part B — PAC-Bayes for Thresholds

$\mathcal{X}_N = \{1, \ldots, N\}$, $\mathcal{Y} = \{0, 1\}$, $\mathcal{H}_N = \{h_t : t \in \{1, \ldots, N+1\}\}$ with $h_t(x) = \mathbf{1}[x \geq t]$.

B.1 VC dimension and point-posterior PAC-Bayes

VCdim = 1. A single point x_0 is shattered: $h_{x_0}(x_0) = 1$ and $h_{x_0+1}(x_0) = 0$ (taking h_{N+1} if $x_0 = N$). For *any* two points $x_1 < x_2$, the labeling $(y_1, y_2) = (1, 0)$ is unachievable: $h_t(x_1) = 1 \Rightarrow t \leq x_1 < x_2 \Rightarrow h_t(x_2) = 1 \neq 0$. Hence $\mathrm{VCdim}(\mathcal{H}_N) = 1$.

KL for a point posterior. P uniform on $N + 1$ thresholds, $Q_S = \delta_{h_{\hat{t}}}$:

$$\mathrm{KL}(Q_S \| P) \;=\; \log \frac{1}{P(h_{\hat{t}})} \;=\; \log(N + 1).$$

PAC-Bayes bound. In the realizable setting, $L_S(Q_S) = L_S(h_{\hat{t}}) = 0$, so

$$L_{\mathcal{D}}(\delta_{h_{\hat{t}}}) \;\leq\; \sqrt{\frac{\log(N + 1) + \log(2n/\delta)}{2(n - 1)}}.$$

Why the $\log(N + 1)$. A VC realizable bound for thresholds gives the much sharper $O(\log(1/\delta)/n)$ rate with *no* $\log N$. The $\log(N + 1)$ in PAC-Bayes is the KL of a point mass against a uniform prior over $N + 1$ atoms: the prior is fixed before seeing the data and is forced to spread its mass across all candidate thresholds, so concentrating onto the empirical winner costs $\log(N + 1)$ nats of KL. This is a property of the *certificate* (specifically: of using a point posterior with a prior that is uniform over an N-dependent set), not of the underlying learning problem.

B.2 Version-space posterior

Version space. If $y_i = 0$, consistency requires $h_t(x_i) = 0$, i.e. $x_i < t$, i.e. $t > x_i$. Hence $t > a(S) := \max_{i:y_i=0} x_i$ (with $a(S) = 0$ if no negatives). If $y_i = 1$, consistency requires $x_i \geq t$, i.e. $t \leq x_i$. Hence $t \leq b(S) := \min_{i:y_i=1} x_i$ (with $b(S) = N+1$ if no positives). So

$$V(S) \;=\; \{t : a(S) < t \leq b(S)\}, \qquad |V(S)| = b(S) - a(S).$$

This set is nonempty in the realizable case since the true threshold lies in it.

KL of Q_V. Q_V uniform on $V(S)$, P uniform on $N + 1$ thresholds, so

$$\mathrm{KL}(Q_V \| P) \;=\; \sum_{t \in V} \frac{1}{|V|} \log \frac{1/|V|}{1/(N + 1)} \;=\; \log \frac{N + 1}{|V(S)|}.$$

Empirical risk. For every $t \in V(S)$, h_t is consistent with S, so $L_S(h_t) = 0$; therefore $L_S(Q_V) = \mathbb{E}_{t \sim Q_V} L_S(h_t) = 0$.

Why Q_V can beat the point posterior. Both achieve $L_S = 0$, but $\mathrm{KL}(Q_V \| P) = \log(N+1) - \log|V(S)|$, smaller than $\log(N+1)$ by exactly $\log|V(S)|$. When the version space is large (typical for small n), this gives a genuinely tighter PAC-Bayes certificate.

B.3 Spreading helps KL, but can hurt true risk

True risk of Q_W. The marginal of $\mathcal{D}$ on $\mathcal{X}_N$ is uniform and labels come from h_τ. For any t,

$$L_{\mathcal{D}}(h_t) = \Pr_x[h_t(x) \neq h_\tau(x)] = \frac{|\{x \in [N] : (x \geq t) \oplus (x \geq \tau)\}|}{N} = \frac{|t - \tau|}{N},$$

since the disagreement set is the integer interval between $\min(t, \tau)$ and $\max(t, \tau) - 1$, of size $|t - \tau|$. Hence

$$\boxed{L_{\mathcal{D}}(Q_W) \;=\; \mathbb{E}_{t \sim Q_W} L_{\mathcal{D}}(h_t) \;=\; \frac{1}{N|W|} \sum_{t \in W} |t - \tau|.}$$

Concrete example. Take $N = 20$, $\tau = 11$, and a one-point sample $S = ((1, 0))$. Then $a(S) = 1$, $b(S) = N+1 = 21$, so $V(S) = \{2, 3, \ldots, 21\}$ with $|V(S)| = 20$ — large.

- $\mathrm{KL}(\delta_{h_\tau} \| P) = \log 21 \approx 3.04$.
- $\mathrm{KL}(Q_{V(S)} \| P) = \log \frac{21}{20} \approx 0.0488$. So $\mathrm{KL}(Q_V \| P) < \mathrm{KL}(\delta_{h_\tau} \| P)$.
- $L_{\mathcal{D}}(\delta_{h_\tau}) = 0$ (h_τ is the true labeller).
- $L_{\mathcal{D}}(Q_V) = \frac{1}{20 \cdot 20} \sum_{t=2}^{21} |t - 11| = \frac{1}{400} (\sum_{j=1}^{9} j + 0 + \sum_{j=1}^{10} j) = \frac{45+0+55}{400} = \frac{100}{400} = 0.25$. So $L_{\mathcal{D}}(Q_V) > L_{\mathcal{D}}(\delta_{h_\tau})$.

No contradiction with PAC-Bayes. PAC-Bayes gives an *upper* bound on $L_{\mathcal{D}}(Q)$, not a monotone equivalence: a smaller KL produces a smaller *bound*, but smaller bound does not imply smaller actual risk. Both certificates correctly upper-bound their respective $L_{\mathcal{D}}$ values; spreading the posterior across the version space gives a tighter *certificate* even when it gives a *worse* posterior, because the posterior is chosen to make $\mathrm{KL} + L_S$ small, not $L_{\mathcal{D}}$ small (which the learner does not see).

B.4 Every fixed prior can be attacked

Pigeonhole. P is a probability distribution, so $\sum_{\tau=2}^{N} P(h_\tau) \leq 1$. Since this is a sum of $N - 1$ nonnegative terms, at least one satisfies

$$P(h_\tau) \;\leq\; \frac{1}{N-1} \qquad \text{for some } \tau \in \{2, \ldots, N\}. \quad (\dagger)$$

Fix this τ.

$\mathcal{D}_\tau$ is realizable. Under $\mathcal{D}_\tau$, $\Pr[(x, y) = (\tau - 1, 0)] = \Pr[(x, y) = (\tau, 1)] = 1/2$. Both points are correctly labelled by h_τ: $h_\tau(\tau - 1) = 0$, $h_\tau(\tau) = 1$. So $L_{\mathcal{D}}(h_\tau) = 0$.

Both support points appear. For $S \sim \mathcal{D}_\tau^n$, let A = event that some $x_i = \tau - 1$ and some $x_j = \tau$. Then

$$\Pr[A^c] \;\leq\; \Pr[\text{all } x_i = \tau - 1] + \Pr[\text{all } x_i = \tau] \;=\; 2 \cdot (1/2)^n \;=\; 2^{1-n},$$

so $\Pr[A] \geq 1 - 2^{1-n}$.

Version space collapses to $\{\tau\}$. On A, $a(S) = \tau - 1$ (the only label-0 value in the support) and $b(S) = \tau$ (the only label-1 value), so by B.2, $V(S) = \{t : \tau - 1 < t \le \tau\} = \{\tau\}$.

Forced posterior. Any posterior Q with $L_S(Q) = 0$ is supported on $V(S) = \{\tau\}$, hence $Q = \delta_{h_\tau}$. Then

$$\mathrm{KL}(Q\|P) \;=\; \log \frac{1}{P(h_\tau)} \;\overset{(\dagger)}{\ge}\; \log(N-1).$$

What this does/does not show.

- *Does show:* For any fixed prior P over $\mathcal{H}_N$, there exists a realizable distribution on which every zero-empirical-error PAC-Bayes certificate must incur $\mathrm{KL} \ge \log(N-1)$. So the *bound* it produces is at least $\sqrt{(\log(N-1) + \log(2n/\delta))/(2(n-1))}$. This is a $\log N$ lower bound on the PAC-Bayes *certificate*.
- *Does not show:* A lower bound on the sample complexity of *learning* thresholds. ERM on thresholds achieves $L_{\mathcal{D}}(\hat{h}) = O(\log(1/\delta)/n)$ uniformly in N (VC dimension 1). The certificate is loose; the learner is fine. The slack lives in the choice to (a) use a *fixed* prior chosen before seeing the data and (b) insist on a *zero-empirical-error* posterior. Either relaxing the $L_S(Q) = 0$ requirement (KL-bound trades off against empirical risk) or allowing a data-dependent prior (e.g. localized PAC-Bayes) can sidestep the bound.

Part C — AI Usage Report

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1. Where I used AI

I used Claude Code in four roles on this assignment:

- **Part A.1 (VC dimension via supports) — proof structure.** I asked AI to confirm the cleanest way to pass from $\Gamma_{\mathcal{H}_k}(n) \le (e^2 dn/k^2)^k$ to the $O(k\log(ed/k))$ VC bound — specifically, which form of the "if $m \le k\log(am/k)$ then $m = O(k\log a)$" lemma to cite. I then verified the constants by hand.
- **Part A.2 (SRM oracle inequality) — bookkeeping of the three statistical costs.** AI helped me lay out the table separating *support cost* $k\log(ed/k)$, *level cost* $\log(1/p_k)$, and *confidence cost* $\log(1/\delta)$, and made me state the class-level SRM theorem cleanly before plugging things in. The actual instantiation (each $p_k\delta$ confidence parameter, the $2\cdot\mathrm{pen}$ factor) I wrote and checked myself.
- **Part A.4 (Validation oracle bound) — independence argument.** I asked AI to be a hostile reader on my proof that conditioning on S_1 makes the candidate list $\{w_1, \dots, w_d\}$ data-independent of S_2. This caught the place where I had originally tried to use a single uniform-convergence bound on $\mathcal{H}$ instead of the much sharper finite-class Hoeffding-with-union-bound on d fixed candidates.
- **Part B (PAC-Bayes for thresholds) — example construction.** For B.3 I sketched the $(N = 20, \tau = 11, S = ((1,0)))$ example by hand, then asked AI to double-check the arithmetic $\sum_{t=2}^{21} |t - 11| = 100$ and the resulting $L_{\mathcal{D}}(Q_V) = 0.25$.

2. AI suggestions I accepted (and why)

- **The product form $\binom{d}{k}(en/k)^k \le (e^2dn/k^2)^k$ for A.1.** AI suggested writing the growth-function bound in this "two factors of the same shape" form so the log-linear fixed-point lemma applies immediately. I accepted because it cleans up the algebra and produces the bound with the right constants.
- **Confidence rescaling $\delta \to p_k\delta$ in A.2.** AI's framing — "each level pays $\log(1/p_k)$ because the union bound assigns confidence $p_k\delta$ to level k" — is exactly the right way to explain the level cost in one sentence. I accepted and put it in the interpretation paragraph.
- **Validation comparison table in A.4.** AI suggested a four-row table comparing penalty-SRM and validation (sample used to fit; sample used to choose; support cost; level cost). I accepted because it makes the trade-off explicit and answers the "what is paid for the separation?" question directly.
- **Distinguish certificate vs learning in B.4.** AI insisted I add a final paragraph separating "PAC-Bayes lower-bounds the certificate" from "this is not a sample complexity lower bound." This is the actual point of the problem; without it the conclusion reads as a paradox.

3. AI suggestions I rejected or substantially modified

- **A.1 lower-bound claim.** AI's first draft asserted $\mathrm{VCdim}(\mathcal{H}_k) \ge k\log(d/k)$ without proof and tried to use it as a "matching lower bound." This is true but nontrivial; I demoted the claim to a sanity-check observation that $\mathrm{VCdim}(\mathcal{H}_k) \ge k$ (which is immediate from $\mathcal{H}_I \subseteq \mathcal{H}_k$) and dropped the unproved matching claim.
- **A.4 single-step union bound.** AI's first attempt at the validation bound tried to run *one* uniform-convergence statement over $\bigcup_k \mathcal{H}_k$ on S_2. This is wrong: S_2 only sees the *finite* list $w_1, \dots, w_d$, not all of $\mathcal{H}$; using the full uniform-convergence bound would gives a worse rate ($\sqrt{d/n}$ via VC) instead of the correct $\sqrt{\log d/n}$. I rewrote to make the conditioning on S_1 explicit and use Hoeffding on d fixed candidates.
- **B.3 example.** AI's first proposed counterexample had $|V(S)|$ small and the KL inequality went the wrong way. I rewrote with $S = ((1,0))$, which leaves $V(S) = \{2, \dots, N+1\}$ (i.e. all but one threshold), making the KL inequality strict and the risk inequality strict.
- **B.4 "$\tau \in \{1, \dots, N+1\}$" claim.** AI's first proof of the pigeonhole step used the pigeonhole over all $N+1$ thresholds to get $P(h_\tau) \le 1/(N+1)$, then tried to construct $\mathcal{D}_\tau$ using $x = \tau - 1, x = \tau$. But this construction fails for $\tau = 1$ (no $x = 0$) and $\tau = N+1$ (no $x = N+1$). I rewrote to restrict $\tau \in \{2, \dots, N\}$ and get the slightly weaker $P(h_\tau) \le 1/(N-1)$, matching the assignment's $\log(N-1)$ bound.

4. How I verified correctness

- **A.1 VC bound.** Verified the algebra step-by-step: $\binom{d}{k} \le (ed/k)^k$ (standard, true for $1 \le k \le d$); Sauer–Shelah for $\mathcal{H}_I$ at $n \ge k$; the log-linear inversion lemma applied to $m \le k\log_2(e^2dm/k^2)$. Cross-checked against Shalev-Shwartz–Ben-David Lemma A.2.
- **A.2 SRM theorem.** Cross-checked the statement of the class-level SRM theorem and the $2 \cdot \mathrm{pen}$ factor against SSBD §7.2. Verified the chain $\varepsilon_k(n,\delta) \to \mathrm{pen}(k,n,\delta) = \varepsilon_k(n, p_k\delta)$ produces the boxed bound by direct substitution.
- **A.3 sample complexity.** Verified $s\log(ed/s) + 2\log s = O(s\log(ed/s))$ for $s \le d$ so absorbing $\log(1/p_s)$ into the leading term is legitimate. Verified the boundary $s\log(d/s) \le d$ algebraically: equivalent to $\log(d/s) \le d/s$, true for $d/s \ge 1$ (which is given since $s \le d$).
- **A.4 validation bound.** Verified the two-step argument $(\star) + (\star\star)$ gives the stated bound

by writing out the two events explicitly and applying the union bound at $\delta/2 + \delta/2 = \delta$. Cross-checked the finite-class step against SSBD §11.2.

- **A.5 runtime.** Verified $(ed/k)^k$ is the upper bound on $\binom{d}{k}$ I used; noted polynomial-in-d regime via $k = O(1)$ and superpolynomial regime via $k = \log d$ by direct exponentiation.
- **B.1 VC=1.** Verified by hand the impossibility of the $(1,0)$ pattern on $x_1 < x_2$ from the definition $h_t(x) = \mathbf{1}[x \geq t]$.
- **B.2 version space.** Verified $a(S) < t \leq b(S)$ from the two consistency conditions $\{y_i = 0 \Rightarrow t > x_i\}$ and $\{y_i = 1 \Rightarrow t \leq x_i\}$. Checked the edge cases (no negatives: $a(S) = 0$ gives $V = \{1, \ldots, b(S)\}$; no positives: $b(S) = N{+}1$ gives $V = \{a(S){+}1, \ldots, N{+}1\}$).
- **B.3 arithmetic.** Computed $\sum_{t=2}^{21} |t - 11| = \sum_{j=1}^{9} j + 0 + \sum_{j=1}^{10} j = 45 + 55 = 100$ by hand; verified $L_{\mathcal{D}}(Q_V) = 100/400 = 0.25$ matches the formula.
- **B.4 failure probability.** Verified $\Pr[A^c] = \Pr[\text{all } (\tau - 1)] + \Pr[\text{all } \tau] = 2 \cdot 2^{-n} = 2^{1-n}$ by enumerating the support of $\mathcal{D}_\tau^n$.

AI workflow updates (5 most recent changes)

1. **"Pigeonhole over the right set" rule.** AI tends to apply pigeonhole over the full set ($\{1, \ldots, N{+}1\}$ here), then build a construction that only works for *interior* indices. I now check, after any pigeonhole step, whether my downstream construction has edge cases. This caught the $\tau \in \{1, N{+}1\}$ issue in B.4.
2. **"Finite-class vs uniform-convergence" prompt.** Before applying uniform convergence over a hypothesis class, I now explicitly ask "is this set fixed independently of the sample I'm evaluating on?" If yes, prefer Hoeffding+union over the (finite) set; if no, pay the VC complexity. This fixed the A.4 validation proof.
3. **Concrete-example-first protocol for PAC-Bayes/KL gaps.** Whenever AI claims an inequality between two information-theoretic quantities (e.g. "KL of Q_V is much smaller than KL of δ"), I require a worked numerical example before trusting it. This is how the B.3 example got written.
4. **Costs-table convention.** I added a rule that every oracle inequality I write must come with a row-by-row explanation of each term in the complexity penalty (support cost, level cost, confidence cost, validation overhead). This produced the interpretation paragraph in A.2 and the comparison table in A.4.
5. **"Reject unproved matching lower bounds."** AI is willing to assert VCdim $\geq \ldots$ matching the upper bound without proof. I now demand either a proof or a citation; if neither, I demote the claim to a weaker sanity check (in A.1: $\mathcal{H}_I \subseteq \mathcal{H}_k \Rightarrow \text{VCdim} \geq k$).

I did not use AI to fabricate any reference; every cited result (SSBD Lemma A.2, SSBD §7.2, §11.2) is one I located and read.